MEDGRID SENTINEL — DISCLOSURE STATEMENT

Anneru IAS Global Energy Hackathon 2026 · Team MedGrid Sentinel — Jackson Sixtus Abetianbe (solo entry) · Brief #13 · Track 2 (Digital Solutions & AI) · Setting B (Non-residential)

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1. WHAT EXISTED BEFORE THE SPRINT (Friday 30 October 2026, 12:00 PM ET)

Nothing of this project. The public repository (github.com/Jae3y/medgrid-sentinel) begins with "Initial commit" at 2026-10-30 17:46 WAT (12:46 PM ET), after the sprint opened. No prior code, designs, data or text was brought in; no prior-work tag was needed.

2. WHAT WAS BUILT DURING THE SPRINT (all visible, timestamped, in git history)

Day 1 (30 Oct, first version, built with ZCode): a deterministic 15-minute microgrid simulation engine (PV, battery, generator, grid, power-quality events, tiered dispatch), Next.js product UI, financial/forecast/sizing modules, documentation and first submission artifacts.
Day 2 (31 Oct, rebuild, built with Claude): engine v2 — measured NASA POWER weather, diesel tank and resupply chain, four dispatch strategies on identical hardware, the Sentinel fuel-runway governor / clinical ride-through reserve / generator cycling, generator transfer-break physics, the Brief #13 Maiduguri 39-day line-outage replay, Monte Carlo resilience with published generator reliability data, a solar forecaster scored on held-out data, full-year 20-year economics with EaaS provider returns, sizing optimizer, fleet screen, SMS/USSD alert channel, offline static export; new UI (Line Outage, Resilience Lab, Fleet & Scale, rebuilt Financials/Brief), 101 automated tests, end-to-end browser walkthrough, regenerated deck, one-page summary, documentation and video scripts.

3. AI TOOLS USED (permitted under Handbook s13/s17 with disclosure)

- ZCode (GLM-based coding agent) — 30 Oct: generated the first version of code, tests, documents and artifacts from the team's written brief.
- Claude Code with Claude Opus 5.5 (Anthropic) — 31 Oct: research assistance, engineering re-design, code, tests, documentation and artifact generation, under the team's direction and review.
- AI was used as a development instrument. Engineering judgements, results and claims were checked against computed outputs and primary sources; every number in the deck, summary and scripts is generated by the engine (results/demo-results.json, `npm run results`).
- No AI-generated footage, imagery or fabricated evidence. The demo and pitch videos are edited from real screen captures of the running application (no generated footage of the product), using an AI-assisted video-editing workflow (open-source tools such as FFmpeg, driven by an AI coding agent) for assembly, captions, highlights and sound design, and a score composed in code (no licensed music). The voice-over is the team's own recorded voice (no synthetic voice). Every figure shown in the videos is generated by the project's engine (results/demo-results.json) and audited against it. Product screenshots are real captures of the running application.
- No confidential, partner or personal information was entered into any AI tool.
- The product itself contains no generative AI: dispatch is deterministic; the forecaster is a least-squares model trained on public data.

4. OPEN-SOURCE CODE AND LIBRARIES (used under their standard licences, unmodified)

Next.js 15, React 19, TypeScript, Tailwind CSS 3.4, Recharts 2.15, Zustand 5, lucide-react, clsx (MIT); Vitest 2, tsx, Playwright (tests/walkthrough); pptxgenjs (deck); ReportLab, pypdf, PyMuPDF, Pillow (PDF/preview tooling). Full list in package.json / package-lock.json.

5. DATASETS

- NASA POWER hourly data (ALLSKY_SFC_SW_DWN irradiance, T2M temperature; community RE), public, retrieved 31 Oct 2026 for Maiduguri 2022–2024 and Lagos, Abuja, Kano, Port Harcourt, Accra, Nairobi, Houston 2024. Raw files in data/raw/nasa-power; conversion script scripts/build-weather-data.py. Source: NASA Langley Research Center POWER Project.
- Published statistics (not datasets) used as model parameters, with sources in docs/DATA_SOURCES.md: NREL/TP-5C00-76553 generator reliability (FTS, MTTF, repair time); World Bank Enterprise Survey outage frequency; NERC Band A tariff; Oct-2026 diesel prices; BNEF storage cost benchmark; WHO et al. "Energizing Health" (2023).
- The 250-bed hospital load profile is ILLUSTRATIVE (built from typical equipment classes), not measured data from any named hospital. No patient data exists anywhere in the project.

6. EXTERNAL API SERVICES

None at runtime. NASA POWER was queried once during development to download data; the application ships the data and runs fully offline in the browser with no keys or servers.

7. THIRD-PARTY ASSETS

None beyond the libraries above. Icons from lucide-react (ISC). No third-party logos or images.

8. CLAIMS DISCIPLINE

Results are simulation outputs, not field measurements. The Maiduguri line outage (22 Sep – 31 Oct 2024) and the December 2022 hospital outage are documented public events used to define the scenario; the hospital itself is a

reference model. No partner, pilot, hospital agreement or customer is claimed.

9. OWNERSHIP

The team retains ownership, subject to the event's judging and promotion agreement (Handbook s13).

Declaration: the statements above are true to the best of the team's knowledge at submission time.

Team representative: Jackson Sixtus Abetianbe